

Multiple Choice (10 marks)

1. In Python 3, which of the following operator in python performs exponential (power) calculation on operands?
 - (a) `**`
 - (b) `//`
 - (c) `is`
 - (d) `not in`
2. Which is the smallest asymptotically?
 - (a) $O(1)$
 - (b) $O(n)$
 - (c) $O(n^2)$
 - (d) $O(\log n)$
3. Which of the following spreadsheet functions would be most useful for detecting improbably high or low values that have become part of a data set as a result of data entry errors?
 - (a) A function that averages numeric values in a column or row
 - (b) A function that counts the values in a column or row
 - (c) A function that rounds a numeric value
 - (d) A function that sorts values in a column or row
4. What is the output of `"abc"[-1]` in Python 3?
 - (a) Error
 - (b) a
 - (c) b
 - (d) c
5. What is the output of `"abc"[::-1]` in Python 3?
 - (a) Error
 - (b) abc
 - (c) cba
 - (d) c

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the sum of the list `l = [1, 2, 3, 4]` is 10.
7. True or False: The value of the Python expression `1 + 3 % 3` is 1.
8. True or False: In Python 3, the expression `b[::2]` retrieves the elements at even indices, resulting in `[11, 15, 19]`.
9. True or False: A large Java program that passes tests may still contain undetected bugs.
10. True or False: The function `join(seq)` in Python 3 checks if all characters in a string are uppercase.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to multiply consecutive numbers of a given list.
12. Write a function to find the maximum difference between available pairs in the given tuple list.

End of Paper